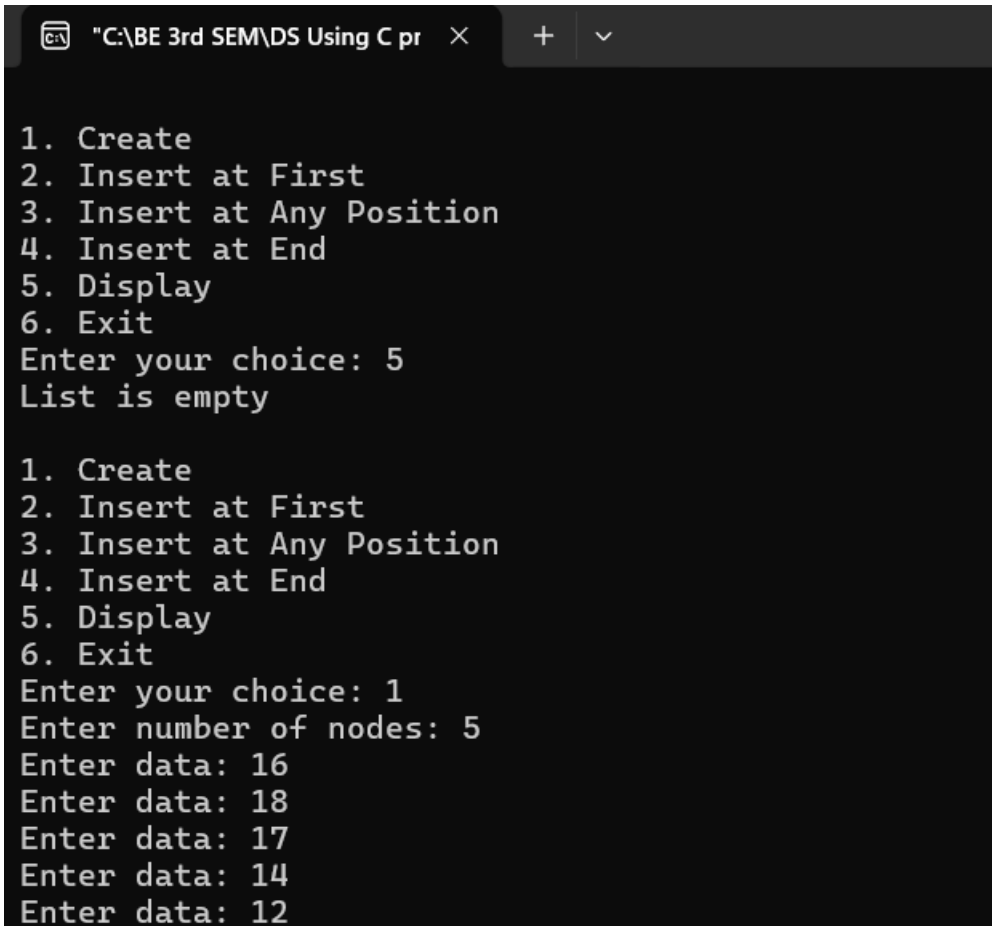
PROGRAM 1: Singly Linked List – Insertion Operations

Aim

To write a C program to implement a singly linked list with the following operations:

1. Create a linked list.
2. Insert a node at the first position.
3. Insert a node at any position.
4. Insert a node at the end.
5. Display the linked list.

OUTPUT:



```
1. Create
2. Insert at First
3. Insert at Any Position
4. Insert at End
5. Display
6. Exit
Enter your choice: 5
Linked List: 16 -> 18 -> 17 -> 14 -> 12 -> NULL

1. Create
2. Insert at First
3. Insert at Any Position
4. Insert at End
5. Display
6. Exit
Enter your choice: 2
Enter data: 10

1. Create
2. Insert at First
3. Insert at Any Position
4. Insert at End
5. Display
6. Exit
Enter your choice: 3
Enter position: 22
Invalid position
```

```
1. Create
2. Insert at First
3. Insert at Any Position
4. Insert at End
5. Display
6. Exit
Enter your choice: 3
Enter position: 4
Enter data: 48

1. Create
2. Insert at First
3. Insert at Any Position
4. Insert at End
5. Display
6. Exit
Enter your choice: 3
Enter position: 7
Enter data: 50

1. Create
2. Insert at First
3. Insert at Any Position
4. Insert at End
5. Display
6. Exit
Enter your choice: 5
Linked List: 10 -> 16 -> 18 -> 48 -> 17 -> 14 -> 50 -> 12 -> NULL
```

```
1. Create
2. Insert at First
3. Insert at Any Position
4. Insert at End
5. Display
6. Exit
Enter your choice: 4
Enter data: 10

1. Create
2. Insert at First
3. Insert at Any Position
4. Insert at End
5. Display
6. Exit
Enter your choice: 5
Linked List: 10 -> 16 -> 18 -> 48 -> 17 -> 14 -> 50 -> 12 -> 10 -> NULL

1. Create
2. Insert at First
3. Insert at Any Position
4. Insert at End
5. Display
6. Exit
Enter your choice: 6
Exiting...

Process returned 0 (0x0)   execution time : 136.665 s
Press any key to continue.
```